Task 6: Neural Network Implementation

Deep Learning Fundamentals & Neural Network Implementation

1. Neural Network Explanation

This task uses an Artificial Neural Network (ANN) built with TensorFlow/Keras to classify breast cancer samples using the scikit-learn Breast Cancer dataset. The dataset contains 30 input features. The data is divided into training and test sets, and StandardScaler is fitted on the training features before being applied to both sets.

The network has two hidden Dense layers with 16 and 8 neurons, both using ReLU activation. The output layer has one neuron with sigmoid activation for binary classification. The model uses the Adam optimizer, binary cross-entropy loss, and accuracy as a metric. It is trained for 50 epochs with a batch size of 16; a portion of the training data is used for validation.

2. Results Comparison

Task 4 reported a test accuracy of 98.2% for standard Logistic Regression. The neural network's test accuracy must be taken from the actual Task 6 run; it is not assumed here. For a strictly fair comparison, both models should be evaluated on the same train-test split.

Model	Test Accuracy	Status
Logistic Regression (Task 4)	98.2% (previous reported result)	Previously reported
Neural Network (Task 6)	Add value from model.evaluate()	Pending execution result

3. Observations

Feature scaling helps neural networks train more effectively when input features have different ranges. ReLU introduces non-linearity in the hidden layers, while sigmoid suits the binary output. Adam updates network weights to reduce the loss.

Training and validation accuracy curves help show learning progress. A persistent gap between them may indicate overfitting; validation loss rising while training loss falls is another warning sign.

The final performance conclusion should be based on actual test metrics and the confusion matrix. Accuracy alone may hide class-specific errors. This is an educational experiment, not a clinically validated diagnostic system.

Note: Task 6 neural-network results were not provided when this report was prepared, so the corresponding accuracy is intentionally marked as pending.